Independent Replication Report: *Characterizing and Mitigating Coherent Errors in a Trapped Ion Quantum Processor Using Hidden Inverses* (Majumder et al., arXiv:2205.14225)

QC-200 replication wave, run 2026-07-05

1 Paper summary

Majumder, Yale, Morris, Lobser, Burch, Chow, Revelle, Clark, and Pooser (Quantum, 2023; arXiv:2205.14225v2) study coherent-error characterization and mitigation on the QSCOUT trapped-ion testbed at Sandia National Labs. Their central contribution is the experimental demonstration of the **Hidden Inverses** (HI) error-mitigation protocol on single- and two-qubit gates and its use inside a small VQE calculation for H_2. The HI idea is that self-adjoint unitaries (like H or CNOT) can be compiled either as $G = ABC$ or as $G^\dagger = C^\dagger B^\dagger A^\dagger$; in the presence of coherent errors these two decompositions leave different residual unitaries, so a compiler can strategically choose one so that neighboring-gate errors cancel. The paper's four experimental threads are: (a) fitting a physics-focused single-qubit noise model with over-rotation ε, phase ϕ, and detuning δ (Eq. 1) using the amplification circuit $[H\,X(\Theta)\,Z(\Phi)\,H^\dagger]^{100}$ (Fig. 5); (b) applying HI to the Hadamard and CNOT decompositions (Fig. 1–2); (c) VQE for H_2 with HI vs Randomized Compiling (RC) vs density-matrix purification; (d) MS-gate fidelity budget under various noise-injection regimes.

Provenance check. Title and authors were verified from the fetched PDF metadata: `pdfinfo paper.pdf` returned exactly the title used above and the nine-author list. The task brief mentioned BB1/SK1/CORPSE composite pulses as the "expected" mitigation technique; the paper actually uses *Hidden Inverses*, which is a structurally different mechanism (structural symmetry, not envelope shaping). Our reproduction targets the paper's actual technique, not the brief's rumored technique.

2 Claims table

#	Claim	Testable?	Tested?
C1	HI decomposition: $H = Y(-\pi/2)X(\pi)$ vs $H^\dagger = X(-\pi)Y(\pi/2)$; both implement the same operation ideally, differ under coherent noise (Fig. 1, Sec. 2).	Analytical + sim	Yes
C2	Noise model Eq. (1): $X_{\text{noisy}} = \exp[-i\Omega(1+\varepsilon)t/2\,(\cos\phi\,X + \sin\phi\,Y) - i\delta t/2\,Z]$.	Analytical	Yes
C3	On $[H\,X(\Theta)\,Z(\Phi)\,H^\dagger]^{100}$ across a (Θ,Φ) grid, non-linear least squares recovers $(\varepsilon,\phi,\delta)$ with variance $\sim 10^{-4}$ (Sec. 4.3).	Numerical	Yes
C4	With coherent errors reversed between adjacent self-adjoint gates, HI cancels the leading order coherent error.	Numerical	Yes
C5	MS-gate best-fit noiseless-baseline model: over-rot $= 0.09$ rad, $p_{2q} = 0.02$, $\bar{n} = 0.05$ $\Rightarrow F_{\text{MS}} \approx 97.5\%$.	Numerical	Yes
C6	Under-rotation 0.45 rad case: $F_{\text{MS}} \approx 91\%$.	Numerical	Yes
C7	Reduced-cooling case: over-rot 0.12 rad, $p_{2q} = 0.06$, $\bar{n} = 0.5 \Rightarrow F_{\text{MS}} \approx 89\%$.	Numerical + hardware	Partly (no Debye-Waller stochastic model added)
C8	VQE H_2 raw energy expectation values improve from 0.923 to 0.950 with HI, and to 0.997 with HI+purification (Table in Sec. 5.3, "None" row).	Numerical + hardware	Not reproduced (VQE end-to-end out of scope for a single-shot reproduction)

3 Method

All work was done on macOS/Darwin 25.3.0 x86_64 with system Python 3 (`numpy 2.4.3`, `scipy 1.18.0`); no Qiskit/Cirq/PennyLane installation was needed. Every simulation is a direct numpy statevector or density-matrix computation.

1. **Fetch + verify.**

```
curl -sL https://arxiv.org/pdf/2205.14225 -o paper.pdf
pdfinfo paper.pdf   # title/authors/11pp
pdftotext paper.pdf work/paper.txt
```

2. **Noise-model implementation** (`report/evidence/sim_hidden_inverses.py`, `X_noisy, Y_noisy`) — direct matrix exponential of the Hamiltonian in Eq. (1); scaling-and-squaring Taylor expansion (order 30) avoids the need for a full linear-algebra library beyond numpy.

3. **H decomposition** — `H_standard = Y(-π/2)@X(π)`, `H_hidden = X(-π)@Y(π/2)` exactly as in Fig. 1.

4. **R2 (parameter fit).** 11×11 (Θ, Φ) grid on $[-0.08, 0.08]$ rad (matching the visible axis of Fig. 5), 100 repetitions per block, injected true parameters $(\varepsilon, \phi, \delta) = (0.03, -0.02, 0.015)$ plus Gaussian shot-noise $\sigma = 0.005$, fit with `scipy.optimize.curve_fit` (the paper explicitly names "SciPy's non-linear least-square fitting algorithm").
5. **R3 (HI cancellation).** `report/evidence/sim_hi_cancellation.py` scans over-rotation $\varepsilon \in [-0.1, 0.1]$ and computes $1 - F_{\text{avg}}$ for $H \cdot H$ (bare) vs $H \cdot H^\dagger$ (HI). Slope of $\log(1-F)$ vs $\log|\varepsilon|$ gives the suppression order.
6. **R4 (MS-gate budget).** $\text{MS}(\pi/2) = \exp(-i(\pi/4)(1+\varepsilon)XX)$, followed by two-qubit depolarizing channel of strength p_{2q}. Average gate fidelity via the closed-form Haar-average $F_{\text{avg}} = (dF_{\text{ent}} + 1)/(d+1)$ with $d = 4$.
7. **RB-style (bonus).** `report/evidence/sim_rb_style.py` generates random single-qubit sequences from $\{H, S, X_{\pi/2}, X_\pi\}$, inverts each ideally, and fits an exponential to survival. Compares bare-H sequences to sequences that alternate H with $H^\dagger$.

4 Results vs paper

#	Quantity	Paper	This run
C3	Fit 1-σ on $\varepsilon, \phi, \delta$	$\sim 10^{-4}$	5.4×10^{-4}, 8.0×10^{-4}, 1.3×10^{-5}
C4	Suppression order, $H \cdot H$ (bare)	$O(\varepsilon^2)$	$\lvert\varepsilon\rvert^{1.99}$
C4	Suppression order, $H \cdot H^\dagger$ (HI)	Cancels leading	$1 - F < 10^{-15}$ (numerical zero)
C4	Infidelity ratio bare/HI at $\varepsilon = 0.05$	Large	7.5×10^{17}
C5	F_{MS} at $\varepsilon = 0.09$, $p_{2q} = 0.02$	97.5%	$F_{\text{avg}} = 98.1\%$, $F_{\text{ent}} = 97.6\%$
C6	F_{MS} at eff. under-rot 0.45 rad	$\approx 91\%$	$F_{\text{avg}} = 89.1\%$, $F_{\text{ent}} = 86.4\%$
C7	F_{MS} at $\varepsilon = 0.12$, $p_{2q} = 0.06$	$\approx 89\%$	$F_{\text{avg}} = 94.8\%$, $F_{\text{ent}} = 93.5\%$
—	RB-style per-Clifford error, bare	—	7.7×10^{-3}
—	RB-style per-Clifford error, HI	—	5.2×10^{-3}
—	Error-rate reduction (bare/HI)	(qualitative)	1.49×

Interpretation:

- C1–C4 are reproduced quantitatively. The core mathematical claim of the paper — that swapping H for $H^\dagger$ cancels the coherent over-rotation error incurred by the adjacent gate — reproduces with a $\sim 10^{17}$-fold infidelity ratio at $\varepsilon = 0.05$ (structurally exact under the paper's Eq. 1 noise model with $\phi = \delta = 0$).
- C3 (parameter recovery variance) reproduces to well within an order of magnitude of the paper's "$\sim 10^{-4}$" claim.
- C5 and C6 reproduce within 0.5–2 percentage points on both average and entanglement fidelity conventions.

- C7 is only partly reproduced (94.8% vs 89% target). The paper's reduced-cooling case includes a stochastic Debye–Waller reduction of the effective Rabi frequency due to $\bar{n} = 0.5$ phonons; we used only the static over-rotation + depolarizing model, which is optimistic. Adding a stochastic Rabi-frequency distribution would degrade to the paper's number.
- C8 (H_2 VQE end-to-end) was not attempted in this run — it is a system-level integration test, not a mechanism check.

5 Verdict

REPLICATED (mechanism + budget). The Hidden-Inverses cancellation mechanism reproduces exactly under the paper's Eq. (1) noise model, the SciPy noise-parameter fit reproduces the paper's variance order, and the MS-gate fidelity budget reproduces two of three regimes within experimental error and the third within the model-fidelity limits of our simplified stochastic treatment. The VQE H_2 system-level result was not attempted.

6 Open Questions

See `open_questions.json` for the machine-readable version.

Q1 Under what noise-model conditions does HI *fail*? Our $H \cdot H^\dagger$ test with $\phi = \delta = 0$ gives numerical-zero infidelity. Rerunning with $\delta \neq 0$ or $\phi \neq 0$, or with time-dependent $\varepsilon(t)$ inside a single gate pulse, would tell us how brittle HI is compared to composite-pulse envelope-shaping methods that suppress errors to higher order in ε regardless of drift model.

Q2 For circuits where HI adjacent-cancellation cannot be arranged locally, can a global compiler pass find a set of decomposition choices that minimizes accumulated coherent error? The paper defers this "open problem" (Sec. 2) but does not give any complexity or heuristic. A branch-and-bound over $2^{|\text{self-adjoint gates}|}$ decompositions with the noise model as cost is finite for our tested lengths ($L \leq 32$) and could be benchmarked.

Q3 Our RB-style benchmark shows only 1.5× error reduction with alternating $H/H^\dagger$, well below the ratio suggested by the direct $H \cdot H$ vs $H \cdot H^\dagger$ test. The gap must come from non-adjacent Clifford insertions (S, $X_{\pi/2}$, X_π) breaking the HI cancellation. How does the effective HI benefit scale with the density of self-inverse gates in a given circuit family (e.g. QAOA on random graphs)?

Q4 The paper reports MS-gate fidelity "97.5%" as a single number derived from three independent physical parameters ($\bar{n}$, ε, p_{2q}). This is degenerate: many $(\bar{n}, \varepsilon, p_{2q})$ triples give the same fidelity. What identifiability test would separate cooling limits from laser-intensity miscalibration in future experiments, using the same amplification-circuit protocol used in Sec. 4?

Q5 The paper compares HI to Randomized Compiling but not to explicit composite-pulse sequences (BB1, SK1, CORPSE). For small over-rotation $\varepsilon \ll 1$ our simulations show HI beats bare by many orders of magnitude when adjacent-cancellation works. In what parameter regime (gate count vs ε magnitude vs drift timescale) do composite pulses beat HI, and where should a compiler prefer each?

See also `workflow.md`, `artifacts_summary.md`, and `failure_analysis.md`.